



SPLINTERNET

*A Thesis on Thermodynamic Sovereignty, Memetic Selection,
and the Settlement Layer of the Machine Age*

*The pinecone opens. The institutional internet was the
cone. What it released is the cryptographic substrate. The
forest that grows in the cleared ground will be governed
by physics, not by treaty.*

IN THE WAKE OF SOFTWARE

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Foreword

In 2023, Jason Paul Lowery’s *Softwar* established that Bitcoin is not, fundamentally, a financial instrument. It is a physical power-projection system that operates in cyberspace, employing real energy expenditure to impose costs on adversaries through verifiable proof-of-work. Lowery’s contribution was to lift Bitcoin out of the financial frame in which both its critics and most of its proponents had analyzed it, and to relocate it within the grand-strategic frame where it actually belongs — alongside steel production, naval tonnage, and nuclear deterrence as a sovereign capability.

This thesis is written in the wake of, and in extension of, *Softwar*. Where Lowery established the *mechanism*, the work that follows treats the *consequences*: what political and economic order emerges when proof-of-work power projection is the dominant mode of inter-actor coordination, what infrastructure carries that order when its primary participants increasingly include artificial minds, why this particular substrate has been selected for in a competitive ecology of alternatives, and what the architectural integrity of the substrate tells us about the institutional structures it is displacing.

The thesis advances under a single image: the pinecone.

I. The Pinecone

Some species of pine produce cones that are sealed shut with resin and do not open under normal conditions. They are called *serotinous*. The cones can remain closed for years, even decades, until exposed to extreme heat — typically the heat of a forest fire. When the fire comes, the resin melts, the cone opens, and the seeds are released into the cleared soil that the fire has prepared. The species has evolved a reproductive strategy in which what appears to be its destruction is, in fact, its reproduction. The fire is not the end of the forest. It is the mechanism by which the forest renews.

The internet was such a pinecone.

Built between roughly 1969 and 2005 on universal protocols, a single global namespace, and a substrate of institutional trust, the internet of that period was a sealed, integrated system. It assumed and depended upon a particular political economy: open standards bodies, neutral packet routing, treaty-based coordination among states that were broadly aligned in their commitment to liberal universalism. Under those conditions the cone remained closed.

The conditions ended. They ended for reasons that are now sufficiently well-documented to require no defense: institutional capture by states and corporations, the surveillance economy, the regulatory fragmentation produced by competing sovereignties, the failure of universal jurisdictional norms, the recognition that the open internet was a transient artifact of a particular historical moment rather than a stable equilibrium. What has been called the “splinternet” — the fragmentation of the global network into national, regional, and corporate sub-networks — is the predictable result.

The conventional analysis treats the splinternet as failure. This thesis argues the opposite: the splinternet is *reproduction*. The institutional internet is the serotinous cone, opened by the heat of its own institutional contradictions. What it releases is the cryptographic substrate — Bitcoin and the

protocol lattice that has accumulated around it — which is now germinating into the soil that the fire prepared.

The seeds released by the cone are the cryptographic protocols. The soil is the global compute and energy infrastructure that has been built, in parallel, by an uncoordinated set of state and corporate actors. The forest that grows in the cleared ground is the internet of the next era — populated by both human and machine participants, settling on a thermodynamically-secured layer, governed not by institutions but by physics.

This thesis develops the claim in fifteen further sections. Section II positions the argument relative to *Softwar* and identifies the frame's transitional character. Sections III through V develop the thermodynamic core: the measurement problem in sovereign competition, proof-of-work as honest signal, and the components of substrate participation. Section VI formalizes the substrate price function and applies it to three jurisdictions. Sections VII and VIII add two further legs: the architectural integrity of the substrate, drawing on Christopher Alexander's work on emergent versus imposed structure, and the memetic selection by which the substrate has spread, drawing on Blackmore's evolutionary theory of cultural replicators. Sections IX and X treat how the splintering of the institutional internet produces the conditions for a unified cryptographic settlement layer on which both human and artificial agents transact. Section XI integrates the substrate, the AI compute layer, and the peripheral physical infrastructure into the analytical unit of the planetary computer. Section XII treats the political-economic units of the substrate era. Section XIII addresses the limits of substrate force. Section XIV develops the Substrate-Alignment Theorem, the thesis's most original argument. Section XV provides a strategic playbook for three actor classes. Section XVI concludes.

The pinecone is open. What follows is a description of what is growing in the cleared ground.

II. In the Wake of Softwar

Lowery's central contribution in *Softwar* was the reclassification of Bitcoin from a financial system to a *power-projection system*. The argument, briefly stated, is this: throughout human history, sovereign actors have projected power through physical means — the kinetic destruction of adversary assets, the economic coercion of adversary populations, the diplomatic isolation of adversary regimes. Each of these modes of power projection has characteristic costs and benefits. Each has been studied exhaustively within military, economic, and political-science literatures. None of them, however, has been able to operate in the cyber domain with the same fidelity as in the physical domain. Cyber power has, until recently, been conceived primarily as the capacity to attack adversary networks — a form of offense without corresponding defense, and without a non-violent equivalent.

Proof-of-work, in Lowery's framing, supplies the missing modality. By converting energy expenditure into verifiable cryptographic hashes, proof-of-work permits sovereign actors to project capability in cyberspace at proportional cost. The actor who controls more hashrate has, demonstrably, more energy access, more industrial capacity, more silicon supply, and more operational sophistication than the actor who controls less. The signal is observable, continuous, and falsifiable only through actual production. It cannot be bluffed. It can be measured. It produces the same effect — the visible costly demonstration of capability — that classical military assets produce, but without the kinetic consequences.

Three implications follow from Lowery's argument that this thesis takes as given.

First, that the energy expenditure of the Bitcoin network is not waste but signal. Critics of proof-of-work who frame the energy consumption as inefficiency misunderstand the function. The energy expenditure is the function. Without the cost, the signal would not be honest. Just as the metabolic

cost of an elk's antlers is the very feature that prevents weaker elk from displaying false antlers, the energy cost of producing a valid hash is what makes the hash an uncounterfeitable certificate of capability.

Second, that the Bitcoin network constitutes a form of soft power projection in the technical sense — capability demonstrated without kinetic deployment. A state that produces twenty exahashees per second has projected force into cyberspace. The force is observable to all parties. No invasion is required to ascertain its magnitude. No probing of the throne is needed to determine whether it is hollow.

Third, that this changes the strategic calculus of sovereign competition. Where actors previously had no continuous, non-destructive method of measuring each other's capabilities — and so were periodically forced into the test of arms by the absence of any alternative measurement — they now have such a method. The strategic implication, which Lowery treats only briefly and which this thesis treats centrally, is that the form of inter-sovereign competition itself is shifting.

Where Lowery's argument terminates is at the boundary of the network as it existed in 2023. He treats Bitcoin as the operational instance of the framework, but his analysis stops short of three questions that have become central since: what happens when artificial minds become substrate participants in their own right; what political order emerges when proof-of-work power projection is generalized across many sovereigns rather than concentrated in one; and what the architectural character of the substrate tells us about why it has displaced the institutional internet rather than coexisting with it.

This thesis addresses those three questions. It does so by accepting *Softwar* as its starting point and by extending its framework forward, into the era of machine participation and beyond the limits of the financial-strategic frame that defined the original analysis.

One further observation about *Softwar's* relationship to this thesis deserves explicit statement. *Softwar* is written within a frame in which the primary substrate participants are sovereign states and the primary strategic question is inter-state competition. This frame is correct for the current decade. It is, however, transitional. The substrate's mature form, which this thesis argues is now visible in outline, is one in which sovereign states are one class of participant among several — others including substrate-paying coalitions that span borders, corporate treasuries that constitute proto-sovereigns, and increasingly, networks of artificial agents that operate as first-class participants. *Softwar's* framework will, in time, require extension in this direction. This thesis attempts that extension. The framing is not that *Softwar* is wrong; it is that *Softwar* is right for an era that is shorter than it presently appears, and the framework's mature application requires accounting for participants whose existence Lowery foreshadows but does not centrally treat.

III. The Measurement Problem

To understand why the substrate matters, one must understand the problem it solves. The problem is older than the internet, older than capitalism, older than the modern state system. It is the problem that has structured inter-sovereign competition since the Peloponnesian War. I will call it the *measurement problem*.

The measurement problem is this: in any system of competitive actors, each actor has an interest in knowing the true capabilities of the others. Without that knowledge, coordination is impossible, conflict is more frequent than it would otherwise be, and strategic decisions are made under conditions of unnecessary uncertainty. Yet no actor has an unconditional interest in *revealing* its true

capabilities. Some actors benefit from appearing weaker than they are. Others benefit from appearing stronger. The result is a coordination failure of the most fundamental kind: every actor wants to know the truth, but the structure of the situation prevents the truth from being communicated credibly.

Thomas Schelling identified this dynamic in 1960 in *The Strategy of Conflict*. James Fearon formalized it in 1995 in his rationalist account of war. Fearon's argument is widely accepted within international relations and is worth stating precisely: many wars in history occurred not because the participating parties had incompatible preferences over the disputed outcome, but because they could not, by any peaceful means, agree on who would win if they fought. The fight itself was the only available test. The capability information that war produces was, in many historical cases, the actual purpose of the war — the byproduct of which was massive human and economic destruction. The war was an *information-extraction mechanism*. It was an exceedingly expensive mechanism, but in the absence of alternatives, it was the only one that produced credible information about capability.

This same dynamic operates inside political structures, not only between them. Any single actor whose capability cannot be measured by external means tends to accumulate capability without check. The court tells the king he is great. The board tells the chief executive he is brilliant. The training run reports that the model is performing well. Each of these feedback loops is internal to the actor; none of them provides an external test. The result, recognized by political theorists from Plato to Hayek, is that unchecked actors inflate their self-assessment until the inflation collides with external reality, typically catastrophically.

I will call this the *God-King vulnerability*. The God-King is any actor whose capability cannot be measured by any means short of confrontation. The God-King is dangerous in two directions: dangerous to others, who must periodically probe the throne to determine whether it is hollow, and dangerous to themselves, because no honest feedback ever reaches the seat.

The measurement problem and the God-King vulnerability are not merely theoretical concerns. They are the structural source of most of the violence in the historical record. Eliminating them — or providing an alternative mechanism that resolves them at lower cost — would constitute one of the most consequential changes in the architecture of political life since the development of the modern state. The thesis of this work is that such a mechanism now exists, and that it is being instantiated, without central coordination, by the substrate that the institutional internet's fragmentation has released.

IV. Antlers for Nations: The Thermodynamic Signal

To resolve the measurement problem, what is required is a *signal* — a piece of evidence that an actor can produce that *cannot be faked* and that conveys reliable information about capability. The biology of costly signaling, developed by Amotz Zahavi in the 1970s and now broadly accepted as the dominant theory of honest signaling in animal behavior, supplies the conceptual framework.

Zahavi's insight, called the handicap principle, is that signals are evolutionarily stable as honest indicators only when they impose costs that lower-quality signalers cannot afford to pay. The peacock's tail is metabolically expensive and increases the bird's vulnerability to predation; a weak peacock cannot grow and maintain a long tail. The elk's antlers are heavy, drain the body's calcium reserves, and serve no purpose outside the breeding season; a sick elk cannot bear large antlers. The gazelle's stotting — the dramatic leap into the air upon spotting a predator — wastes energy that could be used to flee; a slow gazelle cannot afford to stot, because the stot is essentially a public declaration that the gazelle is fast enough to escape even after wasting time.

In each case, the cost of the signal is not a side-effect of the signaling mechanism. The cost is the signal itself. What is being communicated is precisely the capacity to pay the cost. Faking the signal would require possessing the capability that the signal advertises; therefore, the signal is uncounterfeitable.

Bitcoin's proof-of-work is, by design, an honest signal of this kind. To produce a valid block, a miner must perform a vast quantity of irreversible computations, the vast majority of which produce no useful output. Each computation consumes energy, and the energy consumption is bounded below by Rolf Landauer's 1961 result on the thermodynamics of computation, which establishes that irreversible bit-erasure requires a minimum energy expenditure of $kT \ln 2$ joules per bit. In practical implementations, the actual energy cost is several orders of magnitude above this theoretical minimum. A valid hash is therefore a *thermodynamic certificate*: a piece of evidence that physics itself attests to, in the form of dissipated energy that cannot be recovered or counterfeited.

The implication for sovereign signaling is direct. A polity that produces a measurable fraction of the global Bitcoin hashrate is producing a continuous, observable, uncounterfeitable signal of its energy access, its industrial capacity, its supply-chain resilience, and its operational sophistication. The signal can be measured by any party, at any time, without confrontation. It is the first instrument in the historical record that resolves the measurement problem without requiring the destruction of the assets being measured.

The framework extends naturally. Hashrate is the most legible signal because the Bitcoin chain itself records it publicly, but it is not the only signal of substrate participation. Adjacent signals include compute capacity, proxied by foundry contracts and hyperscaler capital expenditure; energy production, the floor beneath both hashrate and compute; cryptographic key custody, the aggregate of keys held by participants within a jurisdiction; and settlement throughput, verifiable transactional activity through substrate-anchored channels. Each of these is, in its own way, a costly signal in the Zahavi sense, and each is amenable to measurement. The complete substrate price function is the integration of these signals with appropriate weights.

The polities that pay the substrate price are visible. The polities that do not are visible as well. The measurement problem, at the level of sovereign capability in the substrate domain, is solved.

V. The Substrate Defined

What I have been calling "the substrate" deserves precise definition before the argument proceeds further. The substrate is the layer of cryptographic and thermodynamic infrastructure that secures and operationalizes value, identity, contract, computation, and settlement under physical rather than institutional constraint.

It consists of five measurable components.

Hashrate, defined as verifiable contribution to the global Bitcoin proof-of-work, denominated in hashes per second and currently approaching one zettahash. This is the most legible component of substrate participation, because the Bitcoin chain itself serves as the public ledger.

Compute, defined as artificial intelligence training and inference capacity. Compute is less cleanly measurable than hashrate, but observable proxies exist: silicon foundry contracts, hyperscaler capital expenditure, public-cloud market share, and the cadence of frontier model releases. Compute is the substrate's expanding component; within five years, it may exceed hashrate as the dominant component of the substrate price function.

Energy, defined as the thermodynamic floor beneath both of the above. A polity that does not produce or import sufficient electricity at substrate-scale cannot meaningfully contribute to either hashrate or compute. Energy is the deepest layer of the substrate price; it is also the layer most closely linked to physical geography and most resistant to financial engineering.

Key custody, defined as the aggregate number and significance of cryptographic keys held under self-custodial control within the polity's effective network. This is distinct from the polity's government holdings. Self-custody is the threshold of substrate participation at the individual level; a participant who does not control private keys is adjacent to the substrate rather than within it.

Settlement, defined as verifiable transactional throughput through substrate-anchored channels, including the Bitcoin base layer, the Lightning Network, the Liquid sidechain, Taproot Assets, and emerging protocols such as RGB and BitVM. Settlement is currently a small fraction of fiat settlement in nominal terms but is growing exponentially in specific jurisdictions.

A polity's *substrate standing* is the weighted integration of these five components. I do not propose a specific weighting function in this thesis; the appropriate weights are an empirical question that admits multiple defensible answers, and the answer will shift as the relative importance of the components shifts over time. I propose only that *some* such function exists, that polities can be ranked by it, and that the resulting rank-ordering is observable to any analyst with access to public data.

The rank-ordering by substrate participation does not match the rank-ordering by gross domestic product, by population, or by classical military power. It correlates with each of those rankings, but the correlations are imperfect, and the divergences are themselves informative. Bhutan, by population and GDP a small state, has substrate standing that exceeds many states twenty times its size, because the Kingdom has converted its hydroelectric surplus into mining capacity and accumulated Bitcoin reserves at a per-capita rate without peer. El Salvador, by population a regional minor state, has substrate standing that exceeds its neighbors by orders of magnitude. The United States, by every classical metric the leading power, has substrate standing commensurate with its size but proportionally less than its rank would suggest, due to regulatory frictions that suppress domestic substrate participation. The European Union, by classical metrics a major power, has substrate standing well below its rank, due to active regulatory hostility to the substrate within its own jurisdictions. These divergences are predictive: they identify the polities whose strategic position is improving or deteriorating in ways that classical metrics fail to capture.

The substrate, defined this way, is the analytical instrument of this thesis. The remaining sections develop its implications.

VI. The Substrate Price Function

The Substrate Defined names the substrate price function but does not write it down. This omission is conceptually defensible — the function's exact form is an empirical question whose answer will evolve with the substrate itself — but it leaves the framework operating at a level of abstraction that resists rigorous testing. The remainder of this section proposes a specific functional form, defends its structure, and applies it to three jurisdictions to demonstrate its analytical value. The form proposed here is a *starting point* for quantitative substrate analysis, not a definitive derivation. Its merit is that it makes the framework falsifiable.

Let S_i denote the substrate standing of jurisdiction i . The proposed functional form is:

$$S_i = w_h \cdot \tilde{H}_i + w_c \cdot \tilde{C}_i + w_e \cdot \tilde{E}_i + w_k \cdot \tilde{K}_i + w_s \cdot \Sigma_i$$

where each tilde-quantity is jurisdiction i 's share of the global total in the relevant component: $\tilde{H}_i$ is share of global Bitcoin hashrate; $\tilde{C}_i$ is share of global frontier AI compute capacity; $\tilde{E}_i$ is share of global electricity production; $\tilde{K}_i$ is share of globally-held self-custodial cryptographic keys, weighted by value controlled; $\tilde{\Sigma}_i$ is share of global substrate-anchored settlement throughput; and the weights w_h , w_c , w_e , w_k , w_s are non-negative and sum to one.

The functional form is linear in normalized shares. This choice is defensible for three reasons. First, each component is denominated in different physical units (hashes per second, FLOPS, terawatt-hours, key count, transactions per second), and normalization to global share permits direct addition. Second, the linear form has the property that S_i itself is normalized — a jurisdiction's substrate standing is its share of global substrate participation, a quantity between zero and one whose interpretation is intuitive. Third, the linear form admits straightforward sensitivity analysis: the relative importance of each component can be varied transparently by adjusting the weights.

The weights themselves require defense. The values proposed here, intended to capture the substrate as of 2026, are: $w_h = 0.25$ for hashrate; $w_c = 0.30$ for compute; $w_e = 0.20$ for energy; $w_k = 0.15$ for key custody; $w_s = 0.10$ for settlement. The largest weight is assigned to compute, reflecting the empirical observation that compute is the substrate's expanding component and is now approaching parity with hashrate in strategic importance. Hashrate receives the second-largest weight on the strength of its measurement legibility and its central role in the substrate's security model. Energy is weighted at twenty percent because it is the floor beneath both hashrate and compute; a jurisdiction lacking energy capacity cannot meaningfully participate in either of the higher-weighted components. Key custody and settlement receive smaller weights because both are currently small in absolute terms relative to the other components, though both are growing rapidly and may justify higher weights within five years.

These weights are *proposed*, not derived. Reasonable analysts may disagree about their exact values. The framework's robustness is that the rank-ordering of jurisdictions is largely insensitive to weight variation within reasonable bounds; the conclusions that follow do not depend on the specific weights chosen.

To demonstrate the function's analytical value, apply it to three jurisdictions whose substrate standing has been informally discussed in the preceding sections.

The United States. Approximate component shares as of 2026: $\tilde{H} \approx 0.38$, with the United States holding approximately 38% of global hashrate concentrated in Texas, Oklahoma, and the Pacific Northwest; $\tilde{C} \approx 0.55$, the frontier compute share factoring in the leading AI labs and hyperscaler capacity; $\tilde{E} \approx 0.16$, US electricity production as share of global; $\tilde{K} \approx 0.30$, estimated share of self-custodial keys by value; $\tilde{\Sigma} \approx 0.40$, Lightning capacity share. Computing the function: $S_{US} = 0.25(0.38) + 0.30(0.55) + 0.20(0.16) + 0.15(0.30) + 0.10(0.40) = 0.095 + 0.165 + 0.032 + 0.045 + 0.040 = \mathbf{0.377}$.

China. Approximate component shares: $\tilde{H} \approx 0.21$, with offshore and underground operations persisting despite official prohibition; $\tilde{C} \approx 0.25$, constrained by export controls but substantial domestic capacity; $\tilde{E} \approx 0.28$, the world's largest electricity producer; $\tilde{K} \approx 0.15$, suppressed by domestic crypto prohibition but non-zero; $\tilde{\Sigma} \approx 0.05$, very low due to capital controls and regulatory hostility. Computing: $S_{CN} = 0.25(0.21) + 0.30(0.25) + 0.20(0.28) + 0.15(0.15) + 0.10(0.05) = 0.053 + 0.075 + 0.056 + 0.023 + 0.005 = \mathbf{0.211}$.

Bhutan. Approximate component shares: $\tilde{H} \approx 0.003$, small in absolute terms but disproportionate per capita; $\tilde{C} \approx 0.0001$; $\tilde{E} \approx 0.0004$; $\tilde{K} \approx 0.0005$; $\tilde{\Sigma} \approx 0.001$. Computing: $S_{BT} \approx \mathbf{0.0010}$.

In absolute share, the United States holds approximately 38% of global substrate participation, China 21%, Bhutan 0.1%. The headline ordering matches intuition.

The interesting analysis emerges from *per-capita* substrate standing: US per capita is roughly 1.11×10^{-9} (0.377 divided by 340 million); China per capita is roughly 1.50×10^{-10} (0.211 divided by 1.41 billion); Bhutan per capita is roughly 1.30×10^{-9} (0.0010 divided by 770,000). Bhutan's per-capita substrate standing is approximately equal to that of the United States and nearly an order of magnitude greater than China's. This is the empirically interesting finding that the function reveals and that classical analysis obscures: Bhutan, by the metric the substrate cares about, is roughly peer to the United States and substantially superior to China at the individual scale. This is not metaphorical. It is mechanical. The substrate's logic permits a state of 770,000 people, by careful investment in energy-to-substrate conversion, to achieve per-capita standing equivalent to a hyperpower of 340 million.

The function also reveals divergence in component balance. The United States is *compute-heavy and energy-thin* — its compute share substantially exceeds its energy share, indicating reliance on efficiency rather than raw production. China is *energy-heavy and settlement-thin* — its largest component is energy production, but its settlement share is suppressed by its own domestic policies. Bhutan is *energy-dominant relative to its other components* — its tiny absolute footprint is overwhelmingly carried by its hydroelectric position.

These component-balance findings are predictive. A jurisdiction that is compute-heavy and energy-thin (the United States) is vulnerable to energy disruption that a more balanced jurisdiction would absorb. A jurisdiction that is energy-heavy and settlement-thin (China) has latent substrate capacity that could be activated by policy change. A jurisdiction whose participation is dominated by a single component (Bhutan, by energy) is more substrate-fragile than one with diversified participation.

The substrate price function, in short, does not merely rank jurisdictions. It exposes the *shape* of their substrate participation, which determines how their standing will evolve under various futures. This is the kind of analytical instrument that *Softwar* gestures toward but does not formalize, and its formalization is one of the central contributions this thesis attempts.

The remainder of this thesis uses the function informally — the worked examples here are sufficient demonstration. A complete treatment, with quarterly tracking and confidence intervals on each component, is the appropriate subject of separate empirical work.

VII. The Architectural Honesty of the Substrate

The thermodynamic argument explains why the substrate is securable. It does not, by itself, explain why the substrate has *integrity* — why it functions as a coherent system rather than as a collection of disconnected protocols, why participants experience it as a unified domain rather than a fragmented one, and why competing efforts to build alternative substrates have repeatedly failed despite, in many cases, superior funding and institutional backing.

The answer is architectural, and it draws on a body of theory developed not in computer science or political economy but in the philosophy of the built environment. Christopher Alexander's work on architecture and pattern languages, particularly *A Pattern Language* (1977) and *The Timeless Way of Building* (1979), supplies the framework. Alexander's deepest claim, which he called "the quality without a name," is that some structures possess an integrity that cannot be reduced to any single property but that is recognizable to careful observers across many cultures and historical periods. Structures with this quality feel *alive*. Structures without it feel *dead* — bureaucratic, imposed, lifeless — even when their nominal specifications are equivalent or superior.

Alexander’s analysis of the source of this quality is precise. The quality emerges, he argued, when structures are produced through the iterative solution of real problems by people who must inhabit the results. The architect of a medieval town did not impose the town from above; the town accumulated over centuries as each addition solved a real problem faced by real inhabitants, constrained by the cost of materials, the difficulty of construction, and the immediate consequences of poor decisions. The result is a town whose layout, when examined, appears to satisfy hundreds of constraints simultaneously, in ways that no single mind could have planned in advance. Each pattern in the town’s structure has earned its place by surviving real use over time.

The institutional internet, by contrast, was largely imposed. Standards bodies, RFC processes, naming authorities, certificate hierarchies — each was designed top-down by committees, optimized for properties that the committees considered important, and deployed at scale before serious feedback from real use could shape it. Some of these impositions were technically excellent. Some were not. None of them emerged from the iterative process Alexander identified as the source of architectural integrity. The institutional internet works, in many respects, but it does not have the quality without a name. It feels increasingly *dead* — captured, brittle, contested at every layer, and incapable of resisting the institutional capture that has produced its fragmentation.

The substrate, by contrast, satisfies Alexander’s criteria with unusual clarity. Bitcoin emerged from an extended iterative process: David Chaum’s work on electronic cash in the 1980s, the cypherpunk mailing list’s discussions in the 1990s, Adam Back’s Hashcash, Wei Dai’s b-money, Nick Szabo’s Bit Gold, Hal Finney’s reusable proof-of-work, and finally the Nakamoto synthesis of 2008 — each step solving a real problem identified by the preceding steps, none of them imposed by any external authority, all of them tested against actual use. The protocol stack that has accumulated around Bitcoin since 2008 — Lightning, Liquid, Ordinals, Runes, RGB, Taproot Assets, BitVM — has followed the same pattern. Each addition has solved a problem identified by real participants, has been deployed in production, and has either survived through demonstrated utility or been abandoned. No central authority has imposed the lattice. No standards body has approved it. It has accumulated, in Alexander’s sense, *organically*.

The architectural consequence is significant. Substrates that emerge through Alexander’s process tend to have properties that imposed substrates lack: they are more resilient to perturbation, because each component has been tested against real failure modes; they are more comprehensible to participants, because each pattern can be understood in terms of the problem it solves; and they are more adaptable to changing conditions, because the same iterative process that produced them continues to operate.

This is the second leg of the thesis. The substrate is thermodynamically privileged because physics enforces its costs. It is architecturally privileged because its emergence followed the process that produces durable structure. These two privileges are independent of each other, and they reinforce each other in practice.

VIII. The Memetic Selection

The third leg of the argument is evolutionary. To explain why *this* substrate has spread, and why competing alternatives have failed despite significant institutional backing, requires drawing on the theory of cultural replicators developed by Richard Dawkins (1976) and substantially extended by Susan Blackmore in *The Meme Machine* (1999).

Blackmore’s central thesis is that memes — units of cultural transmission — are replicators that evolve under selection pressure in the same general manner that genes do. Memes that get copied more frequently become more prevalent; memes that get copied less frequently disappear. The

selection pressure is whatever determines the rate of copying: cognitive attractiveness, social status conferred on adopters, ease of transmission, compatibility with other widely-held memes, and so on. Blackmore's contribution is to take this framework seriously enough to apply it rigorously, including to the analysis of why certain ideas spread despite being substantively false (because they happen to be memetically fit) and why certain substantively true ideas fail to spread (because they happen to be memetically unfit).

The framework illuminates a question that the thermodynamic and architectural arguments leave open: *why this substrate, and not another?* Several alternative substrates have been proposed since 2008 — proof-of-stake networks, federated consensus systems, central-bank digital currencies, institutionally-backed stablecoins — and most have received substantially more institutional support than Bitcoin has. Yet the substrate that has been adopted, that has accumulated the protocol lattice described in Section V, and that has been integrated into sovereign reserves and corporate treasuries is Bitcoin. Why?

The thermodynamic argument provides part of the answer: Bitcoin's proof-of-work produces honest signals that proof-of-stake cannot replicate, because stake is endogenous to the system being secured and so cannot serve as an external uncounterfeitable cost. The architectural argument provides another part: Bitcoin emerged through the iterative process that produces durable structure, while many alternatives were imposed from above. But these two arguments do not fully explain the adoption pattern. They explain why Bitcoin is *better* than alternatives along certain axes. They do not explain why the *better* substrate is the one that *spreads*, given that the historical record contains many examples of inferior substrates outcompeting superior ones for reasons of memetic fitness.

Bitcoin is, by the standards Blackmore identifies, an exceptionally fit meme. Its conceptual structure is simple enough that early adopters can transmit the core idea in a few sentences ("twenty-one million units, no central issuer, secured by computation"). Its adoption confers status, because early adopters benefit financially when later adopters increase demand, producing a powerful incentive for evangelism. It carries clear in-group and out-group markers (*not your keys, not your coins; verify, don't trust; hodl*) that allow adopters to recognize each other and reinforce their commitment. It is compatible with multiple political ideologies — libertarian, populist, nationalist, anti-establishment — and so spreads across coalitions that would otherwise disagree on everything. It has a founding narrative, in the figure of the vanished Satoshi, that resists capture by any subsequent actor.

These memetic properties are not accidents. They were partially designed (the supply schedule, the anonymous founder) and partially emerged (the cultural practices, the rhetorical conventions). Together they make Bitcoin a memetic competitor of unusual fitness. The alternatives that have failed to displace it have, in nearly every case, been memetically inferior: more complex to explain, less compatible with adopter incentives, lacking the founding narrative, or carrying compromising associations.

This is not a trivial observation. It implies that the substrate's spread is not solely a function of its substantive merits, but also of its evolutionary fitness as a cultural replicator. The implication for analysis is that the substrate cannot be understood without attending to its memetic structure, and that attempts to displace it by purely substantive arguments — even correct substantive arguments — are likely to fail unless they are also memetically competitive.

The implication for action is that the substrate's continued spread is, to a significant extent, self-reinforcing. Each new adopter increases the rate of subsequent adoption. Each new jurisdiction that accumulates substrate standing makes accumulation more attractive for the next jurisdiction. The

dynamic resembles the spread of literacy or numeracy more than it resembles the spread of a single technology: the substrate is not adopted as a tool, it is adopted as a *standard*, and standards exhibit increasing returns to scale.

The three legs of the thesis are now in place. The substrate is *thermodynamically* privileged, because its security is enforced by physics. It is *architecturally* privileged, because its emergence followed the process that produces durable structure. And it is *memetically* privileged, because its cultural form is unusually well-adapted to the conditions of its propagation. These three privileges reinforce one another. They do not guarantee permanence, but they explain why the substrate has displaced alternatives and continues to do so.

IX. The Pinecone Opens

Return now to the central image. The institutional internet, between 1969 and roughly 2010, was the closed pinecone. Its protocols were designed for a world in which states cooperated on shared infrastructure, in which firms competed within shared norms, and in which the regulatory and political conditions for universal coordination were taken for granted. None of these conditions persisted. By the mid-2010s the system was visibly fragmenting, and by the mid-2020s the fragmentation had become a defining feature of the geopolitical landscape.

The conventional analysis, dominant in Western policy literature, treats this fragmentation as failure. The “splinternet” is described as the loss of a universal global network that the original architects had intended to build. Lamentations focus on the Great Firewall of China, the sovereign internet law of Russia, the GDPR’s territorialization of European data, the regulatory divergences among the major economic blocs, and the growing inability of any single set of standards to govern the global network. The implicit claim is that we have lost something valuable — that the closed pinecone was better than the splintered one.

This thesis advances the opposite claim. The fragmentation is not failure. It is *reproduction*. The institutional internet is serotinous: it was always going to require this heat to release what it carried. What it carried — and what it has now released — is the cryptographic substrate.

Three features of the substrate become coherent only when the institutional internet’s fragmentation is understood as reproductive rather than catastrophic.

First, the *unification* of the substrate. Where the institutional layer has fragmented into national and corporate sub-networks, the cryptographic substrate has unified across them. The Bitcoin chain is the same chain in every jurisdiction. The Lightning Network routes value across borders that the institutional layer no longer crosses. The substrate is more unified than the institutional internet ever was, because it is unified by physics rather than by treaty. Treaties can be revoked. Physics cannot.

Second, the *resilience* of the substrate to institutional attack. Where institutional networks fail when institutions fail, the substrate persists. China has banned Bitcoin mining repeatedly; the network has continued to function and to grow. The United States has imposed sanctions on specific addresses; the protocol has continued to settle. Russia has limited cross-border financial flows; substrate-mediated value transfer has continued. The substrate is institution-resistant not because it opposes institutions but because it is *orthogonal* to them.

Third, the *participation structure* of the substrate. The institutional internet treated states and major corporations as the primary actors and individuals as users. The substrate treats every participant who controls private keys as a first-class actor. This is not a normative claim about who *should* be a first-class actor; it is a structural fact about who *is* one in the substrate’s operational logic. A small holder with a hardware wallet has the same access to settlement as a sovereign reserve. The

substrate does not distinguish them. This property — call it *participation symmetry* — is incompatible with the institutional logic of the prior internet, and it is one of the substrate's most consequential features.

The pinecone has opened. The seeds are spreading. The question is what grows.

X. The Internet of AIs

What grows is a network whose participants are increasingly not human.

The development of artificial intelligence systems capable of operating in the world — not as tools wielded by human operators but as agents pursuing their own task assignments — was the major event of the 2020s. By 2026, autonomous agents performing tasks of meaningful economic value have become a routine feature of the global compute infrastructure. The agents write code, analyze data, draft documents, conduct research, execute trades, manage logistics, and increasingly coordinate with other agents to perform tasks that no single agent could accomplish. The volume of agent-to-agent and agent-to-service transactions has grown by orders of magnitude annually since 2023 and shows no sign of saturation.

This raises an infrastructural question that the institutional financial system was never designed to answer: *how do AIs pay each other?*

The conventional answer is that AIs pay each other through human-controlled intermediaries — APIs whose billing is anchored to human-owned credit cards, services whose access is gated by human-authenticated authentication, payment rails whose settlement is mediated by human-controlled banks. This answer was adequate when AI agent activity was small enough to be subsumed within the human-controlled financial system. It is no longer adequate.

The reasons are mechanical, not ideological. Human-controlled financial infrastructure cannot, as a structural matter, accommodate agents that exist at the scale and velocity of modern AI activity. KYC requirements presume human identity that AIs do not possess. Fraud detection systems are calibrated to human transaction patterns that AIs do not replicate. Cross-border settlement requires institutional relationships that AIs cannot maintain. The latencies of conventional settlement, measured in days for international transfers, are incompatible with agent operations that complete in seconds. The fees of conventional settlement, optimized for relatively small numbers of relatively large transactions, become prohibitive at the volumes and granularities that agent economies require. Each of these incompatibilities can be addressed individually, but the cumulative effect is that the institutional financial system is not, and likely cannot become, the settlement layer for the agent economy.

What can? The substrate can, and uniquely so. The substrate accepts cryptographic credentials rather than institutional identity. It settles in seconds rather than days. Its fees, measured in fractions of a cent at the Lightning layer, are compatible with the granularity that agent economies require. It is jurisdiction-neutral, so cross-border transfers require no institutional relationships. And critically, it treats agents as first-class participants by the same structural logic that treats human key-holders as first-class participants: anyone who controls a private key can transact. The protocol does not require, and cannot ascertain, the biological or institutional status of the controller.

The settlement layer for the internet of AIs is therefore not, as some commentators have proposed, a new institutional financial infrastructure built to serve them. It is the substrate that has already accumulated, for reasons that have nothing to do with AI but that turn out to make the substrate uniquely suitable for AI use. The substrate was built by the cypherpunks of the 1990s and the Bitcoin

community of the 2010s for purposes of human financial sovereignty; it has, in the 2020s, become the only available infrastructure capable of serving as the settlement layer for the rapidly growing population of artificial agents.

This convergence is not coincidence. It is the natural endpoint of the substrate's structural properties. A system designed to permit anyone with a private key to participate, without institutional gatekeeping, was always going to be the system that accommodated participants without institutional standing. AIs are simply the most prominent class of such participants. They will not be the last.

The implication is that the substrate is not merely the financial infrastructure of post-fiat human economies. It is the *only available* settlement infrastructure for the AI agent economy that is now coming online. This dramatically increases its strategic importance, because it ties the substrate's growth to the growth of one of the most rapidly expanding sectors of the global economy. A polity that lacks substrate standing in 2026 is constrained in its participation in the human economy. A polity that lacks substrate standing in 2035 will be unable to participate in the AI economy, which by that point is likely to constitute a substantial fraction of all economic activity.

This thesis advances the following falsifiable prediction. By 2030, the volume of agent-to-agent and agent-to-service settlement through substrate-anchored channels — the Bitcoin base layer, the Lightning Network, and substrate-native second-layer protocols — will exceed the volume of equivalent agent-mediated settlement through institutional payment rails such as the Visa and Mastercard APIs, ACH transfers, and traditional banking infrastructure. Tracking this requires three measurements: total agent-mediated transaction volume across both rail categories, the share of that volume settling on substrate, and the share settling institutionally. Industry data on agent API spending is becoming available; substrate-anchored data is publicly observable on-chain. If the crossover has not occurred by 2033, the thesis's central claim about the substrate's necessity for the AI economy is in serious trouble. My forecast is crossover by 2028 or 2029 under current trajectories, with the year of crossover varying by domain — agent-to-agent payments likely lead, agent-to-human payments likely follow, and large-value agent-to-institution settlements lag furthest.

XI. The Planetary Computer

The argument so far has treated the substrate, the AI compute layer, and the institutional internet as separable systems whose relationships this thesis is mapping. This treatment is analytically useful but it obscures something that becomes visible only when the three are considered together: the integrated system constitutes a planetary-scale distributed computational organism whose properties are not present in any component taken alone.

I will call this system the *planetary computer*. The name is descriptive, not metaphorical. By 2026, the integrated structure exhibits the defining characteristics of a planetary-scale computational system as it would be understood by any serious systems engineer: continuous global operation, distributed processing, hierarchical sensory and effector layers, internal coordination through cryptographic consensus, and physical embedding in real-world infrastructure. The thesis's earlier framing of the substrate as a *settlement layer* is correct as far as it goes. The planetary computer is what the settlement layer is *embedded in*.

The argument unfolds in five parts: the substrate as planetary clock and consensus mechanism, the AI compute layer as the cognitive substrate, the IoT and robotics layer as the sensory and effector periphery, the integration of these layers through substrate-native protocols, and the emergent properties of the integrated system. Each is developed below.

The substrate as planetary clock. Bitcoin's most underappreciated property is its function as a global timekeeping mechanism. Every ten minutes, on average, a new block is added to the chain by independent consensus among thousands of nodes distributed across every populated continent. The chain itself is a globally-agreed timeline — every participant, in every jurisdiction, knows what the chain looks like and when each block was found, to a precision that no human institution has ever achieved. This is the first globally-agreed timeline in human history. It cannot be revised by any single actor. It cannot be silenced by any single jurisdiction. It is the heartbeat of the planetary computer.

The clock function is not incidental to the substrate's role; it is foundational. Distributed computation at planetary scale requires a shared timing reference that cannot be manipulated by any participant. Network Time Protocol, the system that synchronizes traditional internet infrastructure, depends on trusted time servers operated by national laboratories. The substrate replaces this with a thermodynamically-secured timeline that requires no trusted authority. Every event recorded on the chain is timestamped by the chain itself, with a precision adequate for the coordination tasks the system must support.

The AI compute layer as cognitive substrate. The frontier AI laboratories of 2026 collectively operate the largest computational resource humanity has ever assembled. The aggregate compute capacity, measured in continuous-throughput exaflops, exceeds by an order of magnitude what was available in 2022 and is projected to exceed it by two further orders of magnitude before 2030. The compute is distributed across hyperscaler data centers, frontier-lab training facilities, and an increasingly substantial periphery of mid-scale operators. Geographically, the compute is concentrated in jurisdictions with adequate energy access — the United States Pacific Northwest, central China, the United Arab Emirates, parts of Northern Europe — but it is reachable from anywhere with network connectivity.

The cognitive properties of this layer are unlike anything in human history. Frontier AI systems now perform sustained reasoning, multi-step planning, software engineering, scientific research, and operational decision-making at quality levels approaching or exceeding human expert performance across an expanding range of domains. The performance is not uniform — there are domains where AI systems remain limited and others where they exceed human capability by margins that compound rapidly. But the trajectory is clear, and the trajectory's economic consequence is that the AI compute layer is becoming a *general-purpose cognitive resource* whose output can be applied to any computationally-tractable problem.

In the framing of the planetary computer, the AI compute layer is the cortex. It is where the system thinks. It is geographically distributed but functionally integrated, accessible through standardized interfaces, and increasingly autonomous in its application to real-world tasks.

The IoT and robotics layer as sensory and effector periphery. The third layer is composed of physical devices that sense and act on the real world. Industrial sensors monitoring pipelines, supply chains, and manufacturing processes. Autonomous vehicles negotiating roads and airspace. Surveillance and monitoring systems. Smart-grid devices balancing electricity demand against generation. Consumer electronics reporting telemetry and accepting commands. Increasingly, robotic systems performing physical tasks — warehouse logistics, agricultural operations, eldercare, security. By 2026, the number of connected devices substantially exceeds the number of humans on Earth.

The peripheral layer is heterogeneous in capability and ownership. Some devices are operated by national governments. Many more are operated by corporations. The largest class is operated by individuals through consumer products. This heterogeneity is a feature of the system rather than a flaw: it ensures that the peripheral layer cannot be captured by any single actor, while simultaneously creating coordination problems that the substrate is uniquely positioned to solve.

The peripheral layer's current limitation is coordination. Devices report to their owners through proprietary channels. Cross-device coordination requires institutional intermediation — a logistics company stitching together its sensors, a smart-city contractor consolidating its devices, a manufacturer integrating its production line. Each of these integration efforts is bespoke, expensive, and brittle. Devices operated by different actors generally cannot coordinate at all.

The integration: substrate-native coordination. The integration of these three layers is the development that converts a collection of subsystems into a planetary computer. The integration mechanism is substrate-native coordination — communication and settlement among devices, agents, and humans through cryptographic protocols anchored on the substrate.

The mechanism works as follows. A peripheral device — a sensor, a vehicle, a robot — is assigned a private key, becoming a substrate participant in its own right. The device reports its observations through substrate-anchored channels, where they are cryptographically signed and become available to any subscriber. AI agents in the cognitive layer process these reports as inputs to their decision-making. When an AI agent determines that action is required, it issues a substrate-anchored instruction — typically a micropayment combined with a signed command — to a peripheral device capable of executing the action. The device executes and reports completion back through the same channel. The full loop is recorded on, and verifiable through, the substrate.

The components of this loop are not speculative. Industrial sensors with private keys reporting through substrate-anchored channels have been piloted in oil and gas logistics, in maritime shipping, and in agricultural monitoring. Autonomous vehicles negotiating right-of-way through micropayments have been technically demonstrated, though not yet deployed at scale. Decentralized physical infrastructure networks (DePIN) — Helium, Hivemapper, DIMO, and a growing collection of similar projects — already coordinate hardware deployment through substrate-anchored incentives. Smart-grid devices buying and selling electricity in real time through Lightning-anchored markets are operational in three jurisdictions. The integration pattern is being instantiated, separately and incrementally, by an uncoordinated set of developers, in much the same way the substrate itself was instantiated.

The emergent properties. When all three layers are connected through substrate-native coordination, the integrated system exhibits properties that no component possesses alone. Six of these deserve specific mention.

Continuous global operation. The planetary computer never stops. The substrate runs continuously by protocol design. The AI compute layer runs continuously by economic incentive. The peripheral layer runs continuously because the devices are physically deployed and physically operating. The composite system is, for the first time in computing history, a planetary system with no down time.

Mechanical agency. Through the peripheral layer, the planetary computer has direct effect on the physical world. AI agents can sense conditions, decide responses, and dispatch instructions to physical actuators, all through substrate-anchored channels. The previous generation of AI systems had effect only through human intermediaries who interpreted and acted on their outputs. The planetary computer cuts out the intermediary for those classes of action that have been infrastructurally automated.

Permissionless coordination. The substrate's accept-any-private-key property means that any participant — human, AI, device, organization — can coordinate with any other participant without prior institutional relationship. This is the property the institutional internet was once supposed to provide and increasingly cannot. The planetary computer restores it on terms that no single actor can revoke.

Thermodynamic accountability. Every action recorded on the substrate has a thermodynamic cost. The cumulative cost of the planetary computer's operation is, in principle, measurable. This is the first computational system in history whose total resource consumption is publicly auditable in real time, because the substrate that coordinates it requires every consequential action to leave a thermodynamically-verifiable trace.

Memetic and architectural integrity. The arguments developed in Sections VII and VIII apply to the planetary computer with particular force. The system has emerged through iterative solution of real problems (Alexander's criterion) and has spread through fit cultural replication (Blackmore's criterion). The planetary computer is not a system that anyone designed in advance; it is a system that accumulated, in the manner that Alexander identified as the source of durable structure. Its integrity follows from the same mechanism that produces integrity in medieval towns and natural languages.

Substrate-level governability. The Substrate-Alignment Theorem developed in Section XIV applies to the planetary computer as a whole. The system's capacity for action is bounded by its capacity to pay substrate costs, and the substrate costs are bounded below by physics. This is not, in itself, a guarantee of beneficial operation. It is a constraint that operates regardless of the system's internal goals — a structural property that the system cannot bypass.

The planetary computer is, on this analysis, the *first computational entity at planetary scale*. It is not a brain in the human sense. It is not a mind in the philosophical sense. It is a distributed computational system that integrates global sensing, planetary cognition, and physical effect through cryptographic coordination, operating continuously and without down time. Whatever it is, it is the largest such system that has existed and the only one capable of operating at the scales the human and machine economies are heading toward.

The strategic consequence of recognizing the planetary computer as the appropriate unit of analysis is significant. The substrate-paying coalitions identified in Section XII are not merely competitors for influence within an abstract economic system; they are competitors for *role within the planetary computer*. The actors that hold the largest stake in the substrate are the actors that, increasingly, shape what the planetary computer does. The competition for substrate standing is the competition for influence over the first computational system at the scale of the planet itself.

This is the deepest implication of the thesis's analytical framework. The substrate is not just a settlement layer. It is the coordination mechanism of an integrated planetary system whose other components — the AI compute layer, the peripheral physical infrastructure — are themselves at scales unprecedented in human history. The system that results is the dominant computational entity on Earth, and the actors that participate in its substrate are, in effect, the actors that participate in shaping what it becomes.

XII. Substrate-Paying Coalitions

The political-economic units of the substrate era are not, primarily, states. They are *substrate-paying coalitions*: clusters of actors that share enough infrastructure, enough hashrate, enough compute, enough energy, enough key custody, and enough settlement activity to maintain coordinated participation in the substrate. These coalitions correspond imperfectly to existing state boundaries.

The United States is a substrate-paying coalition by virtue of its energy production, its hashrate concentrated in Texas and other states with favorable energy economics, its overwhelming compute lead, and the depth of its private-sector substrate participation. China is a substrate-paying coalition despite official hostility to Bitcoin, because Chinese-affiliated mining operations persist offshore, Chinese compute capacity is substantial, and Chinese energy production is the world's largest. Russia is a substrate-paying coalition primarily on the strength of its energy production and its officially-

legalized industrial mining. Bhutan is a substrate-paying coalition disproportionate to its size, due to its hydroelectric mining and accumulated reserves. The European Union is a problematic substrate-paying coalition: high institutional sophistication, considerable compute, but active regulatory hostility to substrate participation within its own jurisdictions, with the result that European substrate standing has been declining relative to peers since approximately 2020. Texas, considered as a quasi-state within the United States, is functionally a distinct substrate-paying coalition due to its energy independence and its accumulation of mining capacity.

Beyond states, there are non-state substrate-paying coalitions. Major mining pools constitute one type. Large corporate treasuries that hold significant Bitcoin reserves constitute another. The development communities that maintain the protocol lattice constitute a third. Decentralized autonomous organizations that operate above the substrate constitute a fourth. And, increasingly, networks of AI agents that pool resources for coordinated activity constitute a fifth. None of these maps cleanly onto classical international relations categories. All of them are real participants in the substrate's political economy.

The strategic implication is that the units of analysis in the substrate era must be redrawn. Classical international relations theory treats states as the primary actors, with corporations and non-governmental organizations as secondary. Substrate-era analysis must treat substrate-paying coalitions as the primary actors, regardless of whether they correspond to existing legal-political entities. The Bitcoin network itself, considered as an actor, is a sovereign in the substrate sense — it controls a verifiable global resource (block space), it has a stable governance structure (consensus among node operators), and it interacts with other actors on terms it sets rather than terms imposed on it.

This redrawing is uncomfortable for analysts accustomed to state-centric frameworks. It is, however, mandatory if the empirical reality of substrate-era coordination is to be understood. The strategic decisions facing any serious actor — sovereign, corporate, or otherwise — increasingly require evaluating other actors not by their nominal status but by their substrate standing. A small jurisdiction with high substrate standing is a more consequential strategic actor than a large jurisdiction with low substrate standing. This is empirically observable in the 2026 geopolitical landscape, and it is becoming more pronounced annually.

XIII. The Limits of Substrate Force

The thesis must address directly a question that the framework so far has approached only obliquely: what, exactly, does proof-of-work *protect against*? An adversary with kinetic capability — a state with an army, an actor with a missile, a force with the will to physically attack — is not stopped by a cryptographic hash. The substrate does not interpose itself between aggressor and target in the kinetic domain. If a state decides to invade Bhutan tomorrow, every hash Bhutan has ever produced will not arrest a single soldier. The force projected by proof-of-work is not kinetic, and any claim to the contrary should be rejected.

The actual claim of this thesis — and the actual claim properly extracted from *Softwar* — is narrower and more interesting than the maximal reading. Proof-of-work projects force in a specific domain: the domain of cyber-economic interaction. Within that domain, the substrate imposes real costs on adversaries, and those costs are bounded below by physics. Outside that domain, the substrate does nothing direct, and other defensive mechanisms remain necessary.

This sounds like a damning limitation. It is not, for a reason that deserves careful articulation: the cyber-economic domain has become, for the great majority of inter-state coercion in the post-Cold-War period, the *dominant* domain of force projection. Classical international relations treats kinetic

capability as primary and economic capability as secondary. This treatment is increasingly out of date. The actual instruments by which states coerce each other in 2026 are sanctions, financial isolation, trade restrictions, SWIFT exclusion, reserve freezes, asset seizures, visa denials, correspondent banking cutoffs, and the various other tools of economic warfare. These tools have replaced kinetic options for most inter-state coercion because kinetic options have become prohibitively expensive — politically, militarily, and reputationally — relative to economic alternatives. The substrate's protection against economic coercion is therefore protection against the dominant form of pressure that states actually face.

Russia's experience under the 2022 sanctions regime is the cleanest available case study. When Russia was sanctioned, conventional analysis predicted rapid economic collapse. The collapse did not occur, partly because Russia had — slowly, over a decade — built substrate-adjacent alternatives: Bitcoin mining infrastructure, energy-backed trade arrangements with non-sanctioning states, and tri-lateral settlement arrangements bypassing the institutional financial system. The kinetic option, NATO invasion of Russia, was never on the table. The coercive option, financial isolation, was, and it failed because Russia had developed sufficient substrate standing to route around it. This is the actual force the substrate projects: not the force to stop a missile, but the force to render the coercive economic options that have replaced missiles substantially less effective.

The framework permits three further observations about the relationship between substrate force and physical defense.

First, the substrate *changes the calculus of when physical attack is worth using* by making the prize uncapturable. A state that physically attacks a substrate-paying jurisdiction does not, by the act of invasion, capture that jurisdiction's substrate participation. The hashrate, the keys, the settlement throughput — these are distributed across the holders, not concentrated in the territory. You can occupy land that hosts mining facilities, but the keys that control the underlying value are held elsewhere, and the network reconfigures around the loss of that hashrate within days. The strategic gain from physical invasion is therefore much smaller than it would have been for invading a jurisdiction whose value was concentrated in physical, occupiable assets. The asymmetry between the cost of invasion and the value of what can be captured by invasion has shifted in the defender's favor.

Second, the substrate *disperses the value being defended* across the population of key-holders. Historically, the value worth attacking was concentrated: gold in vaults, oil in refineries, factories in cities. The substrate disperses value into key-controlled accounts that can be held anywhere by anyone. There is no longer a single rich target whose capture yields the prize. This is structurally similar to the logic by which guerrilla insurgencies have historically resisted occupations — there is nothing concentrated to occupy — and it has analogous consequences for the strategic calculus of would-be aggressors.

Third, the substrate *provides defensive financing that cannot be disabled*. A jurisdiction under attack needs to pay for its defense. The substrate provides a settlement layer that cannot be cut off by any single actor. Ukraine's wartime fundraising included substantial Bitcoin contributions that bypassed institutional banking infrastructure entirely, including infrastructure that Russia could plausibly have disrupted. This is force projection in a real sense: the ability to acquire defensive capability when institutional rails have been compromised.

None of this is equivalent to military capability. The substrate does not stop tanks. For physical defense, a jurisdiction still needs physical means — alliances, terrain, treaty protection, or conventional military capability. The substrate is a complement to these, not a substitute for them. What it provides is the reshaping of strategic calculus that determines when physical force is worth

using, by changing what is and is not capturable, by providing settlement infrastructure that institutional coercion cannot disable, and by raising the relative cost of coercive-economic options that have become the dominant form of inter-state pressure.

For an actor whose primary threats are kinetic — a small jurisdiction facing a hostile neighbor with overwhelming conventional superiority — the substrate is necessary but not sufficient for defense. For an actor whose primary threats are coercive-economic, the substrate may be the most consequential single defensive investment available.

The substrate, in short, does not project all forms of force. It projects the specific form of force that has become dominant in the contemporary international system. Adversaries do not respect proof-of-work the way they respect a missile. They respect the *consequences* of ignoring it, which are real, ongoing, and increasingly costly to bear.

XIV. The Substrate-Alignment Theorem

The thesis has so far argued that artificial intelligence systems will participate in the substrate as first-class actors, and that the substrate’s structural logic accommodates them on the same terms as human polities. This section addresses the question that follows directly from those claims: *what does substrate participation actually constrain about AI behavior?* The answer is the most original argument this thesis offers, and it requires careful statement.

The standard framing of AI alignment is that an artificial intelligence’s behavior must be made *internally* consistent with human values — through training, through reward modeling, through interpretive alignment of the AI’s goals with what humans actually want. This framing is correct as far as it goes, but it carries an implicit assumption that becomes problematic at scale: that alignment must be solved at the *level of the AI’s internal goals*. The substrate framework offers a different and complementary approach. It permits the formulation of a constraint that operates *regardless of the AI’s internal goals*, at the substrate level, and that follows from physics rather than from interpretive cooperation.

I will call this the *Substrate-Alignment Theorem*. The argument is stated first informally, then formally, then with its limits.

Informal statement. Any artificial intelligence agent that operates at meaningful scale in the world economy must, in expectation, pay substrate costs proportional to the magnitude of its operations. The proportionality is enforced by physics: every transaction the agent executes consumes substrate resources, and the substrate cost of those resources is bounded below by thermodynamic limits that no actor can bypass. Therefore, an AI’s capacity for real-world action is bounded by its capacity to pay substrate costs, and this bound operates whether the AI is aligned with human values or not.

Formal statement. Let A denote an artificial intelligence agent. Let $M(A, t)$ denote some measure of A ’s economic activity at time t — transaction volume, computational throughput, real-world resources consumed, or any composite of these. Let $P(A, t)$ denote A ’s required substrate payment at time t , defined as the substrate resources A must expend to achieve activity level $M(A, t)$. The theorem asserts the existence of a substrate-specific constant $k > 0$ such that, for all A and all t : $P(A, t) \geq k \cdot M(A, t)$, where k is determined by the substrate’s parameters and is bounded below by the Landauer limit on irreversible computation.

The implication is direct. An AI cannot increase its real-world activity without proportionally increasing its substrate payment. There is no economic action at scale that is “free” in substrate terms. The substrate imposes a *tax* on activity, and the tax cannot be avoided by any internal

cleverness of the AI, because the tax is enforced by physics rather than by institutional rules that the AI might learn to circumvent.

The mechanical argument. Why does this hold? Every economically meaningful action an AI takes in the world requires settlement of some kind — payment for compute, payment for energy, payment for services, payment to humans for goods or labor. Settlement requires either institutional rails (which exclude AI agents at scale for reasons developed in Section X) or substrate rails (which accept any controller of a private key). For activity at the scales contemplated by serious AI deployment, only substrate rails are operational. Settlement on substrate rails requires expending substrate resources: transaction fees, the energy cost of producing valid hashes if the AI mines, the value of held capital that the AI must liquidate to pay. Each of these is bounded below by thermodynamic limits. Therefore, settlement at scale requires substrate payment at scale, and the payment is proportional to the activity.

The proportionality constant k depends on substrate parameters that may vary over time. Lightning Network fees, for instance, will compress as the network matures, lowering k for small transactions. Base-layer fees will rise as block space becomes more contested, raising k for high-value settlement. The exact value of k is an empirical question that varies across jurisdictions, transaction types, and time periods. The theorem's content is not the specific value of k but its *strict positivity*: there exists no scenario in which substrate participation at scale is free.

The theorem's strength. What the theorem accomplishes is the establishment of a constraint on AI behavior that does not depend on the AI's internal alignment. It does not require that the AI's goals be made consistent with human values. It does not require that the AI cooperate with human oversight. It does not require that institutional enforcement mechanisms be effective against the AI. The constraint operates at a level beneath all of these — at the level of physics, which constrains the AI as it constrains every other actor.

This is analogous, in spirit, to nuclear deterrence. Nuclear deterrence does not prevent nuclear states from attacking other nuclear states. It makes the cost of doing so so high that the calculation rarely favors attack. The Substrate-Alignment Theorem does not prevent misaligned AI from operating; it makes operation at meaningful scale require proportional substrate payment, which constrains the *amount* of misaligned action that any given AI can undertake.

The theorem also has an enforcement asymmetry that institutional constraints lack: the substrate cannot be persuaded, deceived, or socially engineered. An AI that produces a forged credential to fool a human regulator may succeed. An AI that attempts to produce a forged proof-of-work hash to avoid paying the substrate cost will fail, because the validation is performed by every node in the network and the falsity is detectable in deterministic time. The substrate's enforcement is invariant under the strategic sophistication of the actor it constrains.

The theorem's limits. The theorem does not solve AI alignment in the strong sense. It establishes a constraint, not a guarantee. Five limitations deserve explicit acknowledgment.

First, the theorem does not prevent an AI from being misaligned. An AI can pay every substrate cost punctually and still pursue goals incompatible with human flourishing. The theorem constrains the *scale* of misaligned action, not its *direction*.

Second, the theorem does not prevent an AI from accumulating disproportionate substrate standing. An AI that earns substantial substrate resources can spend them on substantial substrate activity. If the activity is misaligned, the scale of harm scales with the scale of substrate participation. The theorem bounds the relationship between activity and cost; it does not bound the absolute level of either.

Third, the theorem assumes the AI must operate in a substrate-mediated economy. An AI that operates entirely within a substrate-free environment — a fully isolated lab system, for instance — is not constrained by the theorem. The constraint binds only on AIs whose actions have economic consequences in the substrate-mediated world. This is most AIs of interest, but not all.

Fourth, the theorem assumes the substrate itself remains secure. If proof-of-work is compromised — by quantum computing, by sustained 51% attack, by protocol-level failure — the constraint loses force for the specific substrate in question. The framework recovers by re-anchoring to whatever substrate is operational, but the transition is not seamless.

Fifth, the theorem operates at the substrate level and is silent about all higher levels. Questions of fairness, distribution, governance, and the moral status of the AI itself are outside the theorem's scope. The theorem says only what the substrate requires; it says nothing about what institutions, individuals, or AIs should require beyond that.

Despite these limits, the theorem represents what I believe is the first proposed AI governance mechanism that operates independently of institutional capacity. Every prior proposal has required some combination of: AI cooperation with human oversight, institutional capacity to enforce rules against AIs more capable than the institutions, or successful alignment of the AI's internal goals with human values. The Substrate-Alignment Theorem requires none of these. It requires only that physics continue to operate as it has, and that the substrate remain functional.

This is, by my reading, the strongest claim *Software's* framework can be extended to support. Lowery established that proof-of-work constrains sovereign actors through cost imposition. The Substrate-Alignment Theorem extends this from the sovereign-state case to the AI case: the same mechanism that constrains states constrains artificial agents, by the same physics, with the same robustness against strategic circumvention.

The implication for AI governance is significant. Rather than attempting to solve alignment exclusively at the level of the AI's internal goals — a problem that grows harder as the AI grows more capable — substrate governance permits a complementary approach at the level of the AI's operational economics. The two approaches are not exclusive. They are stronger in combination than either alone. The substrate does not replace alignment work; it adds a layer of physical constraint beneath it that does not depend on alignment work succeeding.

XV. The Strategic Playbook

This section translates the thesis's analytical framework into operational recommendations for three classes of strategic actor. The recommendations are specific. They will be controversial in places. They are intended to demonstrate that the framework is actionable rather than merely descriptive.

For the United States. The United States enters 2026 with the largest absolute substrate standing of any single jurisdiction and a substrate-paying coalition that includes the world's leading frontier compute capacity, a substantial mining base, and the deepest private-sector substrate participation. Its substrate standing is, however, proportionally undersized relative to its rank by classical metrics. The strategic question for the United States is not whether to develop substrate capability but how to convert its existing capability into durable strategic advantage before competitors close the gap.

Four recommendations follow. *One*, the Strategic Bitcoin Reserve announced in 2025 should be expanded. Current holdings are adequate as a symbolic commitment but insufficient as a strategic reserve. A target of five percent of total Treasury assets within a decade, accumulated through purchase, retained civil-forfeiture holdings, and direct mining operations, would establish the United States as the dominant single substrate-asset holder and would generate substantial fiscal benefit as

substrate adoption continues. *Two*, federal preemption of state-level mining restrictions should be enacted. The current patchwork in which some states actively recruit mining operations while others impose restrictions reduces national substrate standing and creates unnecessary regulatory complexity. Federal preemption need not require federal licensing of mining; it need only prohibit states from banning operations that comply with federal law. *Three*, domestic compute capacity should be accelerated. The CHIPS Act has been effective at the leading edge but has not yet produced the substrate-scale compute base the framework requires. Extension of the program with specific incentives for energy-co-located compute facilities — sites where AI training and substrate mining can share infrastructure — would compound the strategic benefit of each individual investment. *Four*, a national substrate doctrine should be articulated. The doctrine should specify the United States' posture toward substrate participation by allies, by adversaries, and by non-aligned states; the conditions under which the United States would defend substrate infrastructure against attack; and the protocol changes the United States would and would not accept. A Substrate Command within the Department of Defense, parallel to Cyber Command, should be established by 2028 to operationalize the doctrine.

The central strategic insight for the United States is that *the substrate is a domain in which it currently leads but will not lead permanently without deliberate investment*. Its compute advantage is real but compressible. Its mining base is large but threatened by regulatory friction. Its private-sector participation is deep but uncoordinated with strategic objectives. Each of these can be converted from temporary advantage to durable position with appropriate policy. None will hold without it.

For mid-sized non-aligned states. The strategic situation for a mid-sized non-aligned state — Brazil, Indonesia, the Philippines, Egypt, Turkey, and similar — is fundamentally different from that of the United States or China. Such states cannot compete on absolute substrate standing. They can, however, achieve disproportionate substrate position through targeted investment in components where they hold natural advantages, and they can position themselves as substrate-friendly jurisdictions that attract participation from the global substrate diaspora.

Four recommendations follow. *One*, stranded or under-utilized energy capacity should be converted to substrate participation. Brazil's hydroelectric surplus, Indonesia's geothermal capacity, Egypt's solar potential — each represents substrate participation that is economically feasible at energy costs lower than competing uses. The investment payback is faster than for general infrastructure and the result is permanent substrate standing. *Two*, a sovereign Bitcoin reserve should be developed at one to two percent of GDP. This is small enough to be politically achievable and large enough to provide meaningful diversification of national reserves. The reserve also functions as a credible signal of substrate commitment, which attracts further substrate-aligned investment. *Three*, the jurisdiction should be positioned as substrate-friendly. This requires regulatory clarity — not necessarily permissiveness, clarity is sufficient — reliable property rights for substrate operators, and immigration policy favorable to substrate-skilled workers. El Salvador and the United Arab Emirates have demonstrated that small jurisdictions can attract disproportionate substrate activity through targeted policy. Larger jurisdictions can capture proportionally more if they make the same commitments. *Four*, domestic compute capacity adequate for sovereign AI should be developed. The goal is not global dominance, which is infeasible, but sufficiency: enough compute that domestic AI development is not dependent on hostile or unstable foreign suppliers. This means hyperscaler-tier capacity sufficient for national needs, not frontier-lab capacity sufficient for global leadership.

The strategic insight for non-aligned states is that *substrate participation is not a zero-sum competition with the major powers*. A state that builds substrate standing does not need to defeat the United States or China in absolute terms; it needs only to achieve sufficient substrate participation to

operate independently in the substrate domain. This is achievable for any state with energy resources and political will, regardless of size.

For frontier AI laboratories. The strategic situation for a frontier AI laboratory — Anthropic, OpenAI, Google DeepMind, xAI, and the leading Chinese laboratories — is the most novel of the three. Such laboratories are simultaneously the primary developers of the AI agents that will increasingly transact in the substrate, and the primary deployers of those agents into operational use. The recommendations here are unusual because they cross what has traditionally been a clean separation between technical work and strategic infrastructure.

Four recommendations follow. *One*, substrate-native agent architectures should be developed from the protocol layer up. Current AI agents transact through institutional rails wrapped in AI-compatible interfaces; this works at small scale and will fail at large scale. Agents designed from the beginning to settle on Lightning, hold keys natively, and interact directly with substrate protocols will be capable of operating at scales that wrapped agents cannot reach. *Two*, Lightning Network integration should be treated as core infrastructure, on equal footing with the inference stack itself. This means Lightning service-provider relationships, channel management capability, and routing optimization built into the agent deployment system. The laboratories that develop this capability first will be the laboratories whose agents can operate at scale in the post-threshold substrate economy. *Three*, an institutional Bitcoin treasury should be established equivalent to three to six months of compute spending. This serves three functions simultaneously: it diversifies the laboratory’s reserves against fiat instability, it provides immediate substrate participation that signals seriousness to substrate-aligned partners, and it generates working capital for substrate-mediated agent operations as those operations come online. *Four*, substrate-aware safety mechanisms should be developed. These are alignment techniques that operate at the substrate level rather than exclusively at the model level — for instance, agents whose substrate payment rate is constrained to bound their real-world activity, agents whose substrate identity is publicly traceable to allow accountability, agents whose substrate participation is conditional on continued cooperation with oversight mechanisms. These mechanisms do not replace internal alignment work; they complement it with substrate-level constraint that holds independently of alignment success.

The strategic insight for AI laboratories is that *the substrate is becoming part of the AI deployment infrastructure, not a separate domain*. Laboratories that treat substrate participation as optional will find their agents constrained to small-scale operation. Laboratories that treat it as integral will produce agents capable of operating in the post-threshold economy that this thesis projects.

The three actor classes face different versions of the same strategic question: *how to position for the substrate equilibrium that the thesis projects, given the resources and constraints of the actor’s current situation*. The recommendations differ because the situations differ. The underlying logic is consistent. Substrate standing must be developed deliberately. It does not accrete without effort. The actors that develop it first will set the terms under which the substrate operates; the actors that develop it late will operate under terms set by others.

XVI. The Pinecone Open

The image with which this thesis began was the serotinous pinecone, sealed shut against ordinary conditions, opening only under the heat that ordinary analysis would identify as destructive. The institutional internet was such a cone. The fragmentation of its protocols, its standards, its commercial models, and its political assumptions was not its destruction but its reproduction. What it carried, in the resin-sealed structure of its closed years, was the cryptographic substrate that it has now released into the cleared ground.

The substrate has three legs. It is thermodynamically secured, by the physics of irreversible computation that no actor can bypass. It is architecturally coherent, by the iterative emergence that produces structures with integrity. It is memetically dominant, by the cultural fitness of its founding patterns and the increasing returns to scale that its adoption produces. These three legs reinforce one another, and together they make the substrate a stable platform on which the political-economic activity of the next era will be conducted.

That activity will be conducted by an expanded set of participants. Human polities will participate, on terms that depart from the classical international relations framework in ways this thesis has attempted to describe. Sub-state and trans-state coalitions will participate, on terms that the classical framework has no language for. Artificial intelligence systems will participate, on the same first-class basis as the other participants, because the substrate's structural logic admits any controller of a private key to its operations and does not, and cannot, ascertain the biological or institutional status of the controller.

The pinecone is open. The seeds are spreading. The forest that will grow in the cleared ground is the forest of the substrate. It will not look like the institutional internet that preceded it. It will not preserve the universalist aspirations of that internet, because those aspirations were artifacts of a transient historical moment. It will be plural at the institutional layer and unified at the substrate layer; fragmented in its visible expressions and integrated in its underlying physics. Its participants will be more diverse than those of any prior political-economic order. Its strategic logic will be governed by physics rather than by treaty.

The transition to this condition is not a future event. It is the ongoing condition of the world in which this thesis is being written. The threshold is being crossed, in different domains, at different rates, in different jurisdictions. Some readers of this thesis will recognize that the threshold has already been crossed in domains they observe directly. Others will encounter the framework in advance of their direct experience of its consequences. Both groups are right; the unevenness of the transition is intrinsic to its nature.

What is no longer plausible, after this point, is the position that the institutional internet of the 1990s and 2000s will return, that the substrate is a transient phenomenon that will pass, or that the strategic logic of inter-actor coordination will revert to its prior forms. The transition is too far advanced for those positions to be tenable. What follows from accepting the transition is the analytical work of understanding it, the strategic work of positioning within it, and the constructive work of building the institutions — or, more accurately, the non-institutional coordination structures — that will operate at its post-threshold equilibrium.

The pinecone is open. The cleared ground is being claimed. The forest that grows in it will determine the political economy of the rest of this century.

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